Sahaj Bhadja

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Summary

Product engineer with **2+ years of experience** building test-automated platforms, AWS-backed services, and release pipelines with strong focus on product quality, regression safety, and operational support for customer-facing systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Designed and optimized multi-tenant data services across **Azure SQL, PostgreSQL, and SQL Server** for **500+ users** and **10K+** records, reducing dashboard latency by **71%** (**1.2s to 350ms**).
- Built production workflows with **FastAPI**, asynchronous job orchestration, and release validation, processing **5K+ monthly jobs** with **99%+** reliability and structured failure isolation.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Engineered backend services for a self-serve grading product handling **10K+ requests/day** using **Flask, MySQL, AWS RDS, and REST APIs**, with concurrent batch execution for **1K+** uploaded files.
- Implemented CI-supported grading, rubric, and usage-tracking workflows for **150+ instructors**, reducing turnaround time by **60%** while improving operational visibility and release confidence.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Built cross-platform product infrastructure with **20+** automated unit/integration suites using **Jenkins, PyTest, and Mockito**, running nightly regression-style validation across Windows, Linux, and macOS.
- Optimized large-scale APIs and AWS-backed ETL flows with **AWS Glue, S3, and OOP patterns**, improving response times by **30%**, pipeline efficiency by **68%**, and reducing technical debt by **20%**.

Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built a **microservice-based resource platform** with **MERN**, WebSockets, and Azure SQL, synchronizing **2M+ records** across locations and supporting **8** product workflows for allocation, billing, and notifications.
- Automated testing and release workflows with **Jenkins, Docker, Jest, Mocha, and Selenium**, improving build validation to **95%+** and creating monitoring-backed delivery loops for faster issue resolution.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built low-latency **Flask** product APIs for booking, pricing, and payment workflows over Firebase, supporting **50K+ records** with sub-**200ms** responses and **40%** faster booking confirmation.
- Designed streaming-style analytics and webhook infrastructure with **Databricks, Spark, and AWS-hosted microservices**, reducing manual refund handling by **70%** and improving deployment coordination across teams.

Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, C++
Product & Backend: Flask, FastAPI, REST APIs, Microservices, OOP, Async Processing, Developer Tooling
Cloud & Infra: AWS (EC2, S3, RDS, Glue), Azure SQL, Docker, CI/CD, Jenkins, GitHub Actions, VM-based Testing
Quality & Ops: PyTest, Mockito, Jest, Mocha, Selenium, Regression Testing, Monitoring, Documentation

Projects

FinNexus Live | *Kubernetes, MongoDB, WebSockets, CI/CD*

- Built a finance-oriented platform for **50K+ users** and **1M+ transactions**, supporting real-time updates, release automation, and secure transaction workflows across a high-volume user base.

Chat2SQL GitHub | *Java, TypeScript, Docker, PostgreSQL*

- Built a developer-facing SQL workspace supporting **10+** databases, enabling schema exploration, structured query execution, and local multi-service workflows across Java and TypeScript modules.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized data processing API on **AWS CodePipeline and ECS/Fargate**, achieving **87%** accuracy with automated CI/CD releases, validation checks, and production-grade reliability.